AKAL PURKH MECHANICAL WORKS

(An ISO 9001-2015 Company)

Ludhiana, Punjab, India - 141010

CERTIFICATE OF INSPECTION (EN10204 3.1)

Certificate No.	HE/2026-2027/GW/465	**Customer Name:**	GWSCORK
Part Description:	Hex Screw M20X60 w/1N & 1PW	**Date:**	6/4/2026
Order No:	2564	**Invoice:**	HE-2345
Lot No.:	C-5327	**Part No.:**	APM R61C
Finish:	ZINC	**Size:**	M20 / **Pitch** 2.5
Qty:	6000 Pcs.	**Length**	60 mm

Chemical Composition (BOLTS)

Material:	19MNB4-HC					**Specification:**		ISO-898-1					
Heat No:	SR-3792/25D03213												
Elements	C	Si	Mn	P	S	Cr	B	Mo	V				
Min	0.150	-	-	-	-	-	-	-	-				
Max	0.400	-	-	0.025	0.025	-	0.0030	-	-				
Actual:	0.300	0.220	0.990	0.020	0.008	0.430	0.0030						

Dimensional Specification (BOLTS)

						Stamp:		APM M20X60 R61C					
Standard:	ISO 4017			**Grade:**		8.8		**Type:**		After Plating			
Specification:	**Across Flat**		**Across Corner**		**Head Thickness**	**Bolt Length**							
Standard:	Min	Max	Min		Min	Max	Min	Max					
Standard:	29.67	30.00	33.53		12.28	12.71	59.40	60.60					
Actual:	29.77	29.97	34.27-34.34		12.40	12.66	59.68	60.38					

General Mechanical Characteristics of BOLTS

			Specification:	ISO-898-1	
Mechanical Properties	**Standard**	**Test Values**	**Mechanical Properties**	**Standard**	**Test Values**
Hardness (HRC)	23-34	28 - 30	**Elongation (%) min.**	≥ 12	14 - 15
Tensile strength (Mpa)	≥ 830	906 - 943	**Impact strength (-20° C) (KV)**	≥ 27	60 - 64
Yield Stress (MPa)	≥ 640	655 - 685	**Torque (Nm) min.**	-	-
Proof Load (kN)	≥ 147	Pass	**Reduction of Area (%) min.**	≥ 52	54 - 55
Decarburization (mm)	≤0.015	Pass			

Dimensional Specification (NUTS)

Lot No.:			**Stamp**		APM-8SB								
Standard:	ISO-4032		**Grade:**		8								
Specification:	**Across Flat**		**Thickness**		**Across Corner**								
Standard:	Min	Max	Min	Max	Min								
Standard:	29.16	30.00	16.90	18.00	32.95								
Actual:	29.76	29.94	17.46	17.88	34.37								

Chemical Composition (NUTS)

Specifications	ISO-898-2						
Material	10B21			**Heat No.**	F260406		
Element	**C**	**Si**	**Mn**	**P**	**S**	**Cr**	**B**
Min	-	-	0.30	-	-	-	
Max	0.580	-	-	0.048	0.058	-	
Actual	0.214	0.100	0.856	0.016	0.004	-	

General Mechanical Characteristics of NUTS

			Specification:	ISO-898-2	
Mechanical Properties	**Standard**	**Test Values**	**Mechanical Properties**	**Standard**	**Test Values**
Rockwell Hardness (HRC)	36 max.	30 - 33	**Proof Loading (KN)**	≥ 225.4	Pass
Threading	M20X2.5P	Pass	**Threading Specification:**	M20X2.5P	6H

Dimensional Specification (Plain Washers)

Standard:	ISO 7091												
Specification:	**Inner Dia**		**Outer Dia**		**Thickness**								
Standard:	Min	Max	Min	Max	Min	Max							
Standard:	22.00	22.52	35.40	37.00	2.40	3.60							
Actual:	22.12	22.28	36.82	36.96	2.78	3.10							

Chemical Composition (Plain Washers)

Material	Mild Steel				**Heat No.**	-	
C	Si	Mn	P	S	Cr	B	
0.240	0.150	0.450	0.042	0.039			
Mechanical	**Standard**			**Test Values**			
Hardness (HV):	100-200			162-172			

Surface Characteristics

					Specification:	ISO 4042	
	Adherence Test		**Coating Thickness**		**ROHS**		
	Standard	**Test Values**	**Standard**	**Test Values**	**Criteria**	**Standard**	**Test Values**
Bolts	**No Peel Off**	Pass	**≥ 8 µm**	12-17	ROHS/REACH COMPLIANCE		OK
Nuts	**No Peel Off**	Pass	**≥ 8 µm**	10-15			
Washer	**No Peel Off**	Pass	**≥ 8 µm**	10-16			

Mechanical Properties	**Standard**	**Test Values**	**Specification:**
Tensile Resistance (KN):	203.0	222 - 231	

Certified that the products have been inspected and tested in accordance with the conditions and requirements of the contract or purchase order, and unless otherwise noted, conform to the specification(s), drawing(s) relevant thereto.

Quality Manager